Linear Algebra | Quiz for §6.1 | November 13, 2026

Name: _____ R. Hammack Score: _____

Directions: Show and explain all work. Put phones away. No calculators, etc.

1. Suppose $T : \mathbb{R}^3 \to \mathbb{R}^2$ is a linear transformation for which:

$$T\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \quad T\left(\begin{bmatrix} 0 \\ -2 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \text{and} \quad T\left(\begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Find $T\left(\begin{bmatrix} 4 \\ 2 \\ 0 \end{bmatrix}\right)$.